Bihar Engineering University, Patna

B.Tech 6th Semester Examination, 2025

Time: 03 Hours
Full Marks: 70

Course: B.Tech
Code: 100606

Subject: Digital Signal Processing

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 *Choose the correct option / answer the following (Any seven questions only)* [2 x 7 = 14]

(a) Non-parametric spectral estimation uses:
(i) Autocorrelation function
(ii) ARMA model
(iii) Data modeling
(iv) Linear prediction

(b) Compute Z-transform of $\delta[n-2]$
(i) Z^{-2}
(ii) 1
(iii) Z^2
(iv) None

(c) Find the Nyquist rate for a continuous-time signal $x(t) = sin(200\pi t) + cos(600\pi t)$
(i) 100 Hz
(ii) 200 Hz
(iii) 600 Hz
(iv) 300 Hz

(d) For a signal $x(n) = \{1, 2, 3\}$, what is the energy of the signal?
(i) 6
(ii) 14
(iii) 12
(iv) 9

(e) For a system described by $y(n) = x(n) + 2x(n-1)$, what is the output for $x(n) = \{1, 2, 3\}$
(i) {1, 4, 7}
(ii) {1, 2, 3}
(iii) {3, 4, 1}
(iv) {2, 4, 6}

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(f) For a system with transfer function $H(z) = 1/(1 - 0.5z^{-1})$, what is the impulse response?
(i) $0.5^n\, u(n)$
(ii) $0.5^{(n-1)}\, u(n-1)$
(iii) $2^n\, u(n)$
(iv) $2^{(n-1)}\, u(n-1)$

(g) The system $H(z) = 1/(1 - 0.5z^{-1})$ is causal if:
(i) ROC is $|z| > 0.5$
(ii) ROC is $|z| < 0.5$
(iii) ROC is $|z| > 1$
(iv) ROC is $|z| < 1$

(h) What is the impulse response duration of an IIR filter?
(i) Finite
(ii) Infinite
(iii) Zero
(iv) One sample

(i) The DFT of x(n) = {1, 2, 3, 4} is:
(i) {10, -2+2j, -2, -2-2j}
(ii) {10, 2, 2, 2}
(iii) {10, -2-2j, -2, -2+2j}
(iv) {10, -2, -2, -2}

(j) The IDFT of X(k) = {3, 0, 0} is:
(i) {1, 1, 1}
(ii) {3, 0, 0}
(iii) {1, 0, 0}
(iv) {0, 0, 1}

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Q.2 (a) Find IDFT of $X[k] = \{4, -2j, 0, 2j\}$. [7]

(b) Solve the following difference equation using the unilateral Z-transform.

$$y(n) - \frac{7}{12}y(n-1) + \frac{1}{12}y(n-2) = x(n) \qquad for\ n \geq 0$$ [7]

with initial conditions and input are

$$y(-1) = 2, y(-2) = 4 \quad \& \quad x(n) = \left(\frac{1}{5}\right)^n u(n)$$

Q.3 (a) Compare the magnitude responses of FIR and IIR filters designed for the same frequencies. [7]

(b) Compute the cross-correlation between $x[n] = \{1,2,3\}$ and $y[n] = \{3,2,1\}$. [7]

Q.4 (a) Explain the effects of finite register length in FIR filter implementations. Discuss quantization of filter coefficients, round-off noise, accumulator overflow on filter stability and frequency response. [7]

(b) Given two sequences $x_1[n] = u[n]$ and $x_2[n] = (-1)^n u[n]$, find their orthogonal representation and calculate their inner product for $n = 0$ *to* 4. [7]

Q.5 (a) A linear shift-invariant system is described by the difference equation [7]

$$y(n) - \frac{3}{4}y(n-1) + \frac{1}{8}y(n-2) = x(n) + x(n-1)$$

With $y(-1) = 0$ *and* $y(-2) = -1$

Find (a) the natural response of the system, (b) the forced response of the system for a unit step input.

(b) Find the linear convolution of the sequences x(n) and h(n) using DFT [7]

$$x(n) = \{1,2\}, and\ h(n) = \{2,1\}$$

Q.6 (a) Given an input $x[n] = cos(0.2\pi n)$ to a system with $y[n] = x[n] - 0.9x[n-1]$, find the output response. [7]

(b) Design a Butterworth digital filter using the bilinear transformation. The specifications of the desired low-pass filter are: With $T = 1sec$ [7]

$$0.9 \leq |H(\omega)| \leq \pi/2$$

$$|H(\omega)| \leq 0.2; \frac{3\pi}{4} \leq \omega \leq \pi$$

Q.7 (a) Apply the FFT radix-2 algorithm to compute the DFT of [7]

$x[n] = \{1, 2, 3, 4, 4, 3, 2, 1\}$

(b) Design a digital IIR low-pass filter with a passband edge at 1000 Hz and a stopband edge at 1500 Hz for a sampling frequency of 5000 Hz. The filter is to have a passband ripple of 0.5 dB and a stopband ripple below 30 dB. Design a Butterworth filter using the bilinear transformation. [7]

Q.8 (a) Design a Chebyshev IIR digital low-pass filter to satisfy the constraints [7]

$$0.707 \leq |H(\omega)| \leq 1$$

$$0 \leq \omega \leq 0.2\pi$$

(b) Given signal power $5\,W$ and noise power $0.5\,W$, find the output SNR improvement after Wiener filtering. [7]

Q.9 Write short notes on *any two* of the following: [7 x 2 = 14]

(a) Properties of IDFT

(b) Sampling theorem

(c) Properties of z-transform

(d) auto-correlation and cross-correlation

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